

BITS & PIECES

HUG CONVENTION

This is the last issue before the HUG national convention to be held in Chicago. I am planning to attend. I look forward to meeting lot of you personally who I have thus far had only telephone or mail correspondence with. I will not be having a booth, nor will I be selling anything. I come just to talk with you and meet you. I hope to see a lot of you there. I will report on the activity in next issue.

* I would like to introduce you to my new **executive secretary**, Penny Dokken. Penny is an accurate hard working individual with excellent typing and clerical skills, and some electronics background to boot! I can't ask for much more. Her duties are diversified around here, from phone answering to typing and data entry. She has been placed in charge of the direct mail advertising campaign, and will be given more duties as she can accept them. She has been with Quikdata for two months now, and so far, so good.

* The new **Magnolia CP/M** has many nice features including the type ahead buffer via interrupt driven keyboard. A new feature added is auto detect of format for the H37 format. If you tell the system you are on H37, it will automatically adjust to 40 or 80 track, SD or DD, and SS and DS operation. This new CP/M supports over 27 different formats!

* I have been told by Ben Wert to beware of the **Vista dual cabinets**. It appears that some of the power supplies are very marginal and may cause problems. Ben verified 60mV ripple on both the 5 and 12 volt supplies which is not acceptable. He also noted that the voltage of the 12V and 5V supplies for dual drives dropped to an unacceptable level. This was causing many problems with the drives which was very hard to diagnose. As I mentioned before, if you are having drive problems, the first thing to verify is the power supply voltages. Just as a side point, I have heard that some of the Heath H77 dual vertical cabinets are experiencing heat build up problems, keep your eye on this if you have the H77.

If you want to hear a good catch 22, Ben purchased his Vista cabinet from Priority One Electronics. After describing all his problems to them with the Vista cabinets, they said they would not take it back, to contact Vista. Vista told him to go back through his dealer, who said contact Vista. . . How is that for a good one!

* I have been told that Burton Hulland has an HDOS version that will work with the Heath H67 hard disk drives, which presently only supports CP/M.

* Pricing on the low profile Z100 series of 16 bit micro computers is tentatively set at \$3,249 for single drive and monochrome graphics. The model with two drives and color graphics is suggested to sell at \$3,999. The all in one model with two drives, green monochrome built in display and monochrome graphics will be priced at \$4,099, all very competitive prices, considering what you are getting.

* The new **Zenith Data Systems' 13" RGB color monitor** which is compatible with the Z100 series

computers will have a suggested retail price of \$699, a very good price considering what you are getting. It has a bandwidth of more than 15 Mhz, and 480 lines resolution. The ZVM-134 will be available by October.

* The **ZT-1 personal information terminal** mentioned a few issues back will have a suggested retail price of \$699. This is a terminal for those wishing to access information services such as Compu Serve and other electronic bulletin board operations. It comes complete with its own video display, built in 300 baud modem and keyboard. With a printer, it can also be used as an electronic typewriter. Telephone numbers, as well as computer account numbers and passwords, need only be entered once, then can be used by selecting the letter of the alphabet from the menu that matches the corresponding number. This information is retained in memory for as long as 6 months with a rechargeable battery built into the terminal.

* Several individuals who have purchased the HS-012 disk drive exerciser have asked about availability of **alignment disks**. These are available from Dysan Corporation. The 80 track version sells for \$50 (206-30-802020), while the 40 track version sells for \$25 (224A 800183). For more information call the toll free number 800-538-8150 to find your nearest dealer. Pirates, these are analog disks and cannot be duplicated! Refer to Technical Forum for cats eye alignment tips.

* There appears to be some confusion in regard to the MX-80 graphics option, GRAFTRAX. Originally it required a parallel port. Now there are several serial port options for the MX-80 available that will handle the speed, the 8145 serial board (in the MX-80) being one of them. Charles McKenna informs me that the UD.DVD from Jim Texeira (Soft Shop) will work with this if option DTR is used, but there may be problems with sub scripts and super scripts. Charles McKenna is investigating this problem. For those interested, contact Charles at (213) 559-0178.

* Fred Pospeschil informs me that Mark Kroska has completed an **upgrade of Tiny Pascal** which includes a general cleanup of the code and fixes several bugs. Mark also added several useful extensions which include;

- 1) 118 sector .DOC file including examples and info on internal memory locations useful to the programmer.
- 2) Source code "include" file capability.
- 3) EOF and I/O result codes handled in a manner similar to Pascal MT+.
- 4) Expanded the "load file" area to 4K, which now supports both loading into the load file area and writing the load file area to disk.
- 5) Random file (sector level) R/W using the MT+ syntax of SEEKREAD and SEEKWRITE.
- 6) Program chaining.
- 7) Program source code may be in both upper and lower case.
- 8) Underscores are now allowable in variable names for clarity.
- 9) An ability to flush the printer buffer without closing the file.
- 10) You can now read and write bytes with any values, not just the standard ASCII set.
- 11) Expanded the number of files which can be opened at any one time. In addition to the program load/chain

channel, you can have the following types of files open at the same time;

- a) sequential read
- b) sequential write
- c) print (LP:)
- d) Load/Store
- e) random sector R/W

There is also more, which I will not cover here. To that end, Mark will mail a copy to any individual, group or club for only \$6 to cover the cost of the disk, postage and packing. The disk also contains programs to easily resize 100K and 400K disk directories without using DUMP and initialize the 8250 UART for the printer (for CTL/P) without having to actually print anything. Contact Mark at 2818 Carolina NE/ Albuquerque, NM 87110.

* Gordon Richards writes "for those of your readers having problems with the Trionyx Z80 CPU board at 2 MHz or 4 MHz, try replacing IC112 on the front panel board. Install a 74LS03 in place of the 7403 that comes with the Heath front panel." Gordon was having problems booting his system and this cleared up his problem. He also said for \$25 or free with 3 diskettes, Trionyx will give the source code for the FWZ80 front panel monitor ROM.

* George Schreyer/ 412 N. Maria Ave/ Redondo Bch, CA 90277 writes in that Steve Haggins can find examples of how to access the system date from Heath UCSD Pascal (H-SCOOP July Request) in USUS Volume 5, 8, or 13. He also wishes to announce that the USUS (UCSD System Users Society) library volumes 10 through 13 are now available. If you are interested in a copy of the catalogs, write George.

* David Orosz had added an ON-ON switch near the top of his Trionyx Z-H8 CPU board for selecting between 2 and 4 Mhz boot on power up. This was reported in last month issue page 12 incorrectly as the Heath Z-H8 CPU card.

* I understand Wayne Green is receiving good response from his project on converting TRS-80 programs to the Heath/Zenith systems. They are aiming to get to CP/M software on soft sector format.

* Charles Bennett sent me some print sample done on an IDS PRISM-80 printer. Really neat! It's a dot matrix printer capable of justify, proportional spacing, various pitches and more. No carriage returns or line feeds are sent when text is justified. It also features subscript and superscript printing with two control code, enhanced print, and COLOR, which unfortunately I can't show here. It prints any color words using a multi color ribbon by backspacing and printing over various colors to get any color wanted. It is quite impressive and although I don't know the exact price, it was quite reasonable for what you get. Oh yes, the lettering is quite good for word processing, despite being a dot matrix printer.

* Well, I guess Percom has had it with their DD controller board, and this didn't surprise me too much either. With their misleading adds, lack of organization, and communications gap within, this was bound to happen sooner or later. They are dropping their Double density H89 controller board and support. By support I mean the software is going to see no more work, cleaning up or revising. What you got is what you get. They may make their source code available for HDOS for those that want to work on it themselves. Dave Heine called and said he was sending his board back and getting a refund. Perhaps that would not be a bad idea in view of the future support. You actually have legal grounds for this, since this product has not lived up to

the advertised claims.

In fact they had many false claims in their advertising. The REMark add stated free HDOS disk driver for card. Brian Hansen told me they wanted an additional \$39.95 for the driver since they claimed the add was not correct. He ended up getting his driver free, but not the source, which is what he "deduced" that the \$39.95 was for. Like I said, no communication.

* John Bales writes in to say he has been running the S & M G/L and payroll packages by Osborne which are written in CBASIC (\$200 each) and is happy with the programs, except for their slow speed. He says it takes about 3 hours to run two small payrolls from transaction entry through printing of checks and registers.

* David Orosz says "I am thoroughly pleased with the Zero-Wait state modification that was performed on my TRIONYX M-H8 memory. After building the bare board I was getting about 39 soft errors while BOOTING at 4 Meg and no errors at 2 Meg. Running in 4 Meg I experienced about 4 errors for every disk I/O operation. This was with the T-H90 motherboard installed. Now -- no errors at all in 2 or 4 Meg!"

David also comments on Jim Benesch's 4 Meg monitor light (June pg 2). He says the ION and MON lights are OFF when in 4 Meg and only the MON light is off when in 2 Meg.

* Bob Branch writes that he is using a "flippy" disk drive with the special "flippy" disks. These disks have index holes punched in two places so when the disk is flipped, the hole is in the proper place. He says he has had trouble with the Magnolia CP/M getting "BDOS ERRORS". His solution was to cover the extra index hole so only one index hole is seen.

* Hank La Barbara/ 3803 Isabel/ Skokie, IL 60067/ (312) 675-5223 (after 6PM) is offering a repair service for most Siemens and Wangco 5" drives. Hank is "one of us", a fellow Heath user. He says he can repair most drives for under \$75, with an approximate turn around time of two weeks, providing parts are available. If the repair will be more costly, or parts are not readily available, the owner will be contacted. If the drive owner decides not to have the repairs made, he will have to pay a \$10 service charge plus return shipping.

The purpose of this offer is to ease the burden of the hobbyist paying top dollars for repairs, and in that light, this offer is only valid for Heath systems using these drives.

* James Wixson informs me that there is a good text available for users of DBASE II, **Dbase User Guide with Applications**, by Adam Green. It is a text on DBII, and is available for \$30 from Adam Green/ Software Banc/ 46 Bianca Rd/ Duxbury, MA 02332/ (617) 641-1235. I plan to get this and will offer feedback on it after I get a chance to read it.

* A note from Don Long of Environmental Control Systems on the PC-12(A) parallel/clock board for the H89. "For those who have a PC-12(A) with the new improved NICAD (yellow in color), R1 which is from 1K to 1.5K-Ohm is overly conservative which can cause the clock to drift when the computer is powered down. Replace R1 with a value around 390 ohms to solve that problem.

* Has anybody experienced problems using the KILL statement in MBASIC (HDOS)? In a very large complex random I/O program I have written, the KILL usually does not work. However if I do the same KILL from command mode, no problems. Heath software

consultants are convinced there is no problem with it, but I have been over my logic many times and AM convinced there is a problem with it under certain conditions. Anyone else experience anything similar?

* George Duke has had problems with his Microsoft COBOL causing the disk drive to hang after every compile. Again, Heath software consultants are not familiar with this problem. They did say that they have heard of problems of running Microsoft products with the Trionyx memory expansion board, "It has been known to cause problems while compiling or link editing." Any feedback on this?

* George also wrote in expressing his appreciation of HDOS and programs such as Newline and Software Toolworks products and others. He says the company he works for purchased some Apple's because they were cheap. They are having a very difficult time getting proper support, and more problems with the sparse documentation. He also mentions that the programs are very over priced for what you are getting, and very poor quality.

* With the rapid technology of Hard disk drives, and with more companies coming out with them for the Heath systems, I am a bit leary of suggesting anyone to jump into them at this time. It may be safe to hold off for awhile. Besides, I have yet to find an application where my two 8" DD DS disk system (total 2.5 megs) is not sufficient. However, as I mentioned last month, I feel the best buy for the money is the CDR systems hard disk drive.

* I have been getting questions as how to boot off the Tandon 400K drives using the UltiMeth DKH17V2 driver. You simply set it up, INITIALize and SYSGEN to the Tandon, and then you can BOOT off of it either by physically changing the drive to SY0:, or by placing in the "A" register of the H8 computer the drive # and pressing GO, or by B "X" on the H89, where X is the drive you want to boot from.

NEW STUFF

SOFTWARE RELATED

I have taken the time and money to send several NEW STUFF blurps back to vendors who did not include a phone number, as I said they would no longer be published in this newsletter. Starting next issue, no phone number, no print, no sending back to vendors!

* From Jerry Phelps/ 6013 Innes Trace Rd/ Louisville, KY 40222/ (502) 425-4765 comes a new game called **LADDERS**. This is a real time .ABS game under HDOS for the H89 or H8/H19/H17 with H8-4 card and H19, a graphic game in animation. There are 5 floors connected by ladders and a vertical lift which continuously moves up and/or down that is used to move between floors. The player controls a man who may walk, jump or move between floors on the ladders or vertical lift. There are two BAD GUYS who will try to catch your man, who are smart enough to use the ladders, but unable to use the vertical lift. In the upper left corner of the screen is a prisoner facing a rocket, and a flame which slowly advances toward the rocket. Once the flame ignites the rocket it destroys the prisoner and the game is ended.

There are 12 spots scattered throughout the 5 floors. The object is to step once on all 12 spots before all your men are destroyed or the prisoner is blown up. Much more! Supplied on 5" hard sector 40 track, requires 40K memory. Price is \$18.95 which includes delivery in the continental US.

* **SFDVD** is an HDOS 2.0 device driver allowing the H19 screen to be used for data input and display. It allows the H19 screen form access to user programs written in BH BASIC, MBASIC, FORTRAN-80 and Assembly language. With SFDVD, data collection and file update systems can be made more "user friendly". User written application programs need only be concerned with reading and writing data variables. Many more features. Requires an H8/H19 or H89 with 48K RAM and HDOS 2.0. Comes on a 5" hard sector disk with 12 page manual. Available for \$19.95 (plus \$2 shipping & handling) from **Generic Software/** POB 1154/ Troy, MI 48099/ (313) 879-6903 or from most Heathkit Centers.

* From **Evryware/** POB 60802/ Sunnyvale, CA 94088/ toll free (800) 547-5995 (orders) and (408) 738-1614 (questions) has some new software available. **EXTERMINATOR** is a fast action game that pits the player against the anthropods of Yargon. Shoot remote controlled bullets or hop into the Star-Copter to exterminate the deadly anthropods. Plenty of graphics, animation and variety as the action gets fierce. But watch out for the mysterious swamp creatures. Price is \$19.50, requires 48K and HDOS or CP/M.

HIS MAJESTY'S SHIP IMPETUOUS, is another interactive fiction series, taking you back in time to take command of a King's ship. You must contend with the responsibilities of command, the temptation of treasure, and Napoleon's French fleet. Price is \$21.50. Requires 48K, two drives and MBASIC. Both HDOS and CP/M versions available. Both of these products are available on 5" hard or soft sector at most Heath/Zenith retailers or from Evryware.

* Another from **SOFTWARE TOOLWORKS/** 14478 Glorietta Dr/ Sherman Oaks, CA 91423/ (213) 986-4885 is **AUTODIFF**. It is a software package for detecting and reporting differences in CP/M files. It can locate all changes in a new version of a document or program source file, listing the differences or producing a copy of the file with all changes marked. Built-in filters allow AUTODIFF to optionally ignore non-printing characters, and to display control characters on output. Available for \$29.95 on 5" disks, or 8" standard CP/M format. Add \$2 postage and handling.

* From **Interactive Micro Systems/** POB 21007/ Columbus, OH 43221/ (616) 363-1534 comes **KEY-WIZ**, a multi-key search utility. KEY-WIZ was designed to quickly search a file of "index cards" and find all entries which match a profile of keywords, which may be specified along with logical operators such as AND, OR and NOT to provide a very selective search criteria. Included with KEY-WIZ is **SORT-WIZ**, a utility to provide a way to sort the data files used with KEY-WIZ. Available in HDOS .ABS or CPM .COM (2.2.03) format on 5" hard sector disk. Can be used on H8 or H89 with 48K. Price is \$19.95. Specify operating system when ordering.

* A good flyer to have around for a multitude of software products is available from **SUNFLOWER SOFTWARE/** 13915 Midland Dr/ Shawnee, KS 66216/ (913) 631-1333. Their products have been mentioned many times in back issues of this newsletter, some good stuff! Be sure to write for their later flyers. They now have out **S-BASIC** for HDOS, with a CP/M version to shortly follow. S-BASIC is a structured BASIC language which retains the merits of BASIC while replacing its antiquated control structures with modern ones compatible with the philosophy of structured programming. It is also a translator which translates that language into BH EXT BASIC, or MBASIC. Price is \$49.95

Also just released is **GENERAL LEDGER** under HDOS or CP/M for \$99.95. This new G/L by Sunflower is written in 8080 assembler language for speed and compactness. Requires 48K, and supplied in dual format HDOS and CP/M 5" hard sector disk, at least 1 drive, 80 column terminal and 80 column printer. Will hold over 330 accounts and 1300 transactions with 48K memory, and over 800 accounts with over 3000 transactions with 64K (HDOS version). The CP/M version will do about 25% better!

Many features including; 13 user defined options, command completion, auto report generating upon book closing, removal of last transaction, optional password protection, multiple accounting system, "general" and "specific" journal posting, and more.

* This item of interest to the present owners, or future owners of **COMMAND-8**, the voice recognition system for Heath Computers from MI-8/ 822 E. County Rd 30/ Ft. Collins, CO 80525/ (303) 669-4116. Command-8 which was written up in H-SCOOP previously, sells for \$124.95. This new software will enable you to write programs in MBASIC to simplify the use of your COMMAND-8 system. You can use MBASIC to link other programs to COMMAND-8, such as the VB.DVD (V-8 voice synthesizer device driver), BSR home control interface and more. A very handy feature is the ability to SAVE word vocabularies on disk to be called when needed, eliminating the need to train COMMAND-8 each time you power up. Another nice feature is the ability to replace or retrain a single word in the vocabulary without retraining the entire vocabulary of words. Better memory utilization is also now possible with this software.

Several sample programs demonstrating these new features are included on the disk, along with full doc. The **EXTENDED APPLICATION SOFTWARE** package comes on standard 5" hard sector disk and is available from MI-8 for \$29.95, which includes shipping and handling.

* From Hank La Barbara/ 3803 Isabel/ Skokie, IL 60076/ (312) 675-5223 comes a new device driver, **FK.DVD**. Oh no, not another device driver, you are saying. This one is unique! Now you can mount all your disks, or load MBASIC and a program with one keystroke, and program the special function keys to do all the typing you normally do. You could, for instance, have one key for "CAT" and one for "MOUNT SY1:", or a "COPY" key. You could even program a key to call up a program.

FK.DVD is a device driver allowing the top row of special function keys on the H89 or H19 terminal to be programmed with commonly used commands, or any other type of keyboard input up to 60 characters per key. When the selected key is pressed, the preprogrammed text is put into the type-ahead buffer as if the operator had typed it from the keyboard! Keys can also be programmed with the HDOS SET command.

This program may be used interactively with other programs and the keys programmed accordingly. The driver may be placed in a passive state and reactivated on command, or locked out until the next reboot, allowing the driver to be disabled when running programs that may already use these special function keys. Price is \$25.00 which includes shipping & handling within continental US. Requires HDOS 2.0 and H19 or H89.

* **SKETCH.ABS** is a color graphics editor for the HA83/HA89-3 color graphics boards. It operates in graphics 2 mode of the TMS-9918a VDP and allows the

user to define sprites and move them. Using the keypad of the H19/H89, lines may be drawn and erased, Polygons may be defined, scaled, rotated, translated, duplicated, and dragged across the screen. Also includes a circle generator and curve drawer. Areas of the screen may be cleared, or the on/off colors redefined. There is a paint function that will paint any enclosed area or shape. The picture may then be saved and later retrieved from disk. These disk files may then be used in other programs to initialize the contents of VRAM and the screen to the desired state without having to redefine everything in the program.

POLAR.ABS is a demo program included that creates spirograph type images on the color monitor by drawing various functions in polar coordinates. **ENGINE.ABS** is another demo that everyone enjoys watching. It is a cutaway view of a piston engine that accelerates and decelerates with sparks, explosions, and little people jumping up and down all over the place. Requirements are HDOS 1.6 or later with a 48K system, H8/H19 or H89 with the HA83 or HA89-3 color boards. Distributed on 5" hard sector disk, priced at \$35 which includes all programs and doc. To order contact Mark Lucas of **COLORWORKS/** 5337 E. Bellevue Ave/ Tucson, AZ 85712/ (602) 325-0096. Stay tuned to H-SCOOP, as there is more coming in the future from **COLORWORKS**.

HARDWARE RELATED

* If you own an H89, you'll want to write to **FBE Research Co, Inc/** POB 68234/ Seattle, WA 98168/ (206) 246-9815 (eves) for their flyers. They now have the drivers for the NEC8023 printer, and the new EPSON GRAFTRX PLUS machines. They also sell the NEC for \$480, which has many features including graphics, proportional print, friction and sprocket feed, and more. Delivery for the printer is currently about 60 days.

Their **H89UTI** Utility card, contains a real time clock, parallel interface, high speed 8231A or AM9511A math chip, two serial ports, and battery backup. There are many options, write for pricing details.

Their **H89RAM** is the 16K Expansion RAM for the H89 for \$64.95. Write for the flyer which describes these plus their H89CTI series parallel printer interfaces for the H89 and H8CTI for the H8, Graphics software for the Epson MX-100 or GRAFTRAX MX-80, printer drivers and more.

* Also mentioned in here several times before was **W.I.S.E./** 422 Third St/ Baraboo, WI 53913/ (608) 356-9432. Their products include their serial/ parallel /RTC for the H89, with an adapter which allows you to get an additional serial port (for a total of 3) from the board, an H89 extender board, Centronics cables, to mention some. The Centronics parallel cable is 6' long and uses a parallel port of their board. In the center of the cable is a small 'in line' board which does all the signal conditioning to make the signals Centronics compatible. The HDOS demo disk has a complete driver for a nongraphics parallel driver, and if you want CP/M, they will add the Centronics driver to your CP/M Bios for a nominal charge. Write them for a complete catalog or for more info.

* **MPI/** 4426 S. Century Dr/ SLC, UT 84107/ (801) 263-3081 (ask for John) has graphic extensions for the Z89 and Z90 computers. They plan similar support for the new "Z" machines. The **AP-PAK** graphics package provides extensive graphics printing capabilities for the Z89 and 90. Easy to install, the AP-PAK allows the MPI printer to do what no other printer can do. In addition to an unlimited number of character fonts, variable type sizes, and precision dot addressable graphics, the MPI-

Zenith combo makes it possible to print anything that can be stored or displayed on the CRT. It also allows the definition of graphic characters, including letters such as Old English and Hebrew in positive or negative form, to be printed, which cannot be displayed.

Software is included allowing the user to design and create new fonts. The AP-PAK is operable on the MPI 88G and 99G printers, both logic seeking bidirectional printing at speeds of 100 cps. The AP-PAK is available off the shelf for \$55. The 99G printer is a 9 wire dot matrix printer selling for \$849, while the 88G is a seven wire dot matrix printer selling for \$749.

* If you are interested in a complete **point of sale** package for your H/Z89, including all software and electronic cash register, contact **Xtra Soft, Inc/ 4910 Roman Dr/ Louisville, KY 40291/ (502) 499-1533**. They also have an inventory management system and mail base system.

OFFERS

I make all monthly software (H-Sxxx series) offers available on an as-is basis, in other words, no money back policy if not satisfied, mainly because it's too easy to copy discs. You take your chances. At my prices, you really can't go wrong. I don't think I've let anyone down in the past, and will continue to do my best not to. HS-000 to 099 are HDOS offers, while HS-101 to 150 are CP/M offers. All OFFERS supplied on 5" hard sector disks, unless otherwise indicated.

The following is the current offer list available through **QUIKDATA, INC.** As policy, this listing will be printed quarterly.

HS-XXX OFFERS

HS-001 \$10 General Utilities & Toys
 HS-002 \$10 Word Processors & Editors
 HS-003 \$10 H19 Graphics & Printer pic I
 HS-004 \$10 H19 Graphics & Printer pic II
 HS-005 \$10 Labels, Random Files, Plant
 HS-006 \$10 Challenger
 HS-007 \$10 HDOS 2.0 Patches & doc files
 HS-008 \$10 Quikdata Selectric & misc drivers
 HS-009 \$10 Misc Basic Programs I
 HS-010 \$06 Multi Key Sort Routines
 HS-011 \$08 From the Uni of Houston
 HS-012 \$18 Disk drive exerciser routines
 HS-013 \$10 Multi Dup & Dump

The complete description of these (except HS-012 appeared in June) appeared in April 1982 issue. If you did not receive this issue, write for a description.

PROJECT ARTICLES FOR PARALLEL PORT--WRITE FOR CATALOG

Included in this series is A/D/A converter, 32 channel on/off controller, 600 Baud LA-36 mod, Telephone computer interface and more.

PROFESSIONAL SOFTWARE

Spellbinder word processor and office management package. The best word processor around, makes use of H89/ H19 special function keys. This supports true proportional spacing, and is easy to use. It's what this newsletter is done with. Available in 5" hard sector only. Lists for \$495, my price is \$285.

DBASE II data base system by Ashton-Tate, \$495, compare at \$700. The easily programmable complete data base system for mailing lists, business systems, inventory, or whatever you want, that runs under CP/M.

Available in 5" H/Z format, or 8" (56K required). Has the most features of any dbase program on the market. Handles files of up to 65000 records, sorts and indexes on ANY field and multiple fields, full programmability (IF THEN ELSE, DO WHILE etc.). Perhaps this program along with **Spellbinder** is all the businessman or serious hobbyist will ever need. 30 day trial period.

DDUMP by elektrokonsult in Norway for \$29.95. This and the following CP/M utilities by elektrokonsult are quite unique in that they work on ANY single user CP/M 2.X system with ANY diskette size and format, and many hard disks. All are menu driven and easy to use. **DDUMP** makes it possible to examine and patch any byte on any sector addressed by track and logical sector number - or addressed by allocation block number, both in HEX and ASCII format. May be used to study how CP/M allocates disk storage, to examine or repair a damaged disk and to recover lost data.

DTEST by elektrokonsult for \$29.95. This tests a disk for bad spots and locks them out from later use by CP/M. All errors are reported to printer and/or console.

DDUP by elektrokonsult for \$29.95 is a disk duplicator independent of disk controller, drive, disk size and format. It only requires that source and destination disks are of the same format and density. It will also replace bad sectors from the source with blank sectors on the destination, enabling you to automatically recover (what can be recovered) damaged files.

DKH17V2.DVD by Ultimeth Corp for \$40. Used with Tandon drives in a single density system to get 400K per 100-4 drive. Many exceptional features such as settable step times and motor control idle time. Great for 40 TK drives just for the features. For HDOS 2.0 only.

DKDDFLP.DVD for use with the Magnolia H89 DD board to allow HDOS to be used with it. Excellent HDOS driver for a CP/M product! \$40.

BIOS-80 for CP/M 2.2.03 for \$35 (updates from 2.2.02 CP/M version \$10 + original disk). Simple BIOS replacement which retains all BIOS 2.2.03 features plus allows you to use anything from 100K to 400K drives with it single density for 392K storage per TM-100-4 drive.

HARDWARE RELATED ITEMS

MAGNOLIA H89 DD SETUP. One of the best ways to go to get double density for your H89. Complete with CP/M cables and all needed. Supports 4-8" drives AND 4-5" drives in addition to your H17 controller! \$495.

TANDON TM-100-4 drives. \$365. Double side, double track 96 TPI disk drive.

TANDON TM-100-2 drives. \$290. Double side, single track drive.

TANDON TM-100-1 drives. \$225. Single side 40 track drive just like your Siemens.

TANDON 848-2 8" Thinline double sided disk drive for \$495.

SINGLE 5" CABINET (Horizontal) with power supply for 5" drives, \$85.

DUAL 5" CABINET (Heath WH17) with power supply for 5" drives, side by side horizontal, \$170.

MISCELLANEOUS ITEMS

Book set **Your Fortune in the Microcomputer Business**, discussed in June H-Scoop. Two volume set is priced at \$24.95. Volume 1 is "Getting Started", and Volume 2 is "Growth, Survival and Success". Individual volumes are \$15.95. This is the best and most useful book set I have

seen in this area.

TERMS: All prices are cash (credit card add 3%) and includes shipping and handling for UPS delivery (the larger items), or Mail (small software items). For large items shipped by mail (only if UPS does not service your area, or you have POB, etc) add \$5 extra per item over \$200. WI residents add 5% tax.

I also offer a **disk conversion** service. I can upload or download disk varieties from 40 track SS 5" to 8" DD DS Magnolia format, and standard SD SS CP/M IBM 3740 format and most anything in between.

I have separate catalogs, brochures, descriptions and price lists available for most of these products. If you desire further info on any item(s) let me know.

This month's offer is the first CP/M offer I have available, **HS-101 BIGTYPE & BUSINESS FILES** for \$10, another example of good low cost software. Part of this disk is a take off of HS-011 "From the University of Houston, Victoria campus" and has the file **BIGFONT.BAS** converted to run under CP/M's MBASIC interpreter, just as the HDOS version ran. It was thrown in mainly for comparison, and for those who do not have HS-011, the HDOS version. This program produces large letters on the screen to display a message or whatever. The unique thing about these letters is they are formed just as if a human hand were printing them.

In addition, the file has been changed and modified giving it additional features, and compiled to a direct executable .COM file. All source and associated files are included, so you can change and re-compile if you desire. The major change was the data statements in the interpreter version to store the message. This is now a text file, which can be created with any text editor which is then called from the main program. It is presently set up to access this file, **BIGTEXT** on the B: drive. Applications are many, including leaving messages, or as a sales tool in a store, especially a computer store when the system is not being used. Very unique and interesting. Many thanks from Bruce Bennett of Mountain Software for these files.

The business files were contributed by Associated Business Computer Group of Milwaukee, Wisconsin. These alone are comparable with programs selling for many times the price of this disk. The programs are both MBASIC interpreter programs; **AMORTIZE.BAS** and **DEPREC.BAS**. **AMORTIZE** is a program which inputs the user for loan amount, interest, term, etc, and will calculate the amortization schedule of the loan, mortgage, or what have you, and can either list the output to the console or the LST device. It will calculate monthly payment if desired and give you your dates, payment, beginning balance, interest payment, principal payment, ending balance, interest to date, and principal to date - very complete.

The other, **DEPREC** is a depreciation schedule showing the depreciation of equipment over a period of x years - very good for income taxes. This one only outputs to printer. Full disk with only 3K remaining.

REQUESTS

* Morris Yeaman/ POB 4455/ Houston, TX 77210 and several others are trying to locate a program such as **CLIST** which will recognize both the 40 track SS and the 80 track DS drives. Anybody heard of such a program? He wishes to be able to catalog both 40 and 80 track disks, and sort both alphabetically and by extensions the files, to allow him to locate duplicate programs.

* Rick Indiano/ 11 Terry Ln/ Rochester, NY 14624 of **RHUG** (Rochester HUG) is interested in exchanging newsletters, software and ideas with any HUG groups.

* Neil Stone/ Rt. 4, Box 101/ Thief River Falls, MN 56701/ (218) 681-1251 requests hearing from anyone who has successfully interfaced an Olivetti ET 221 Electronic Typewriter to a Heath H8 computer via either the H8-2 parallel or the H8-4 serial ports using the CP/M operating system.

* Ed Ranson/ 35020 45th Ave S/ Auburn, WA 98002 has the following requests: He wants to know how to change the printer port and configuration using **MBASIC**. What he wants to do is change from a TI810 on port 340 to a Diablo 630 on port 320. I'm not sure about the Heath CP/M but the Magnolia can easily be changed in the DOS using the **STAT** command. The actual printer port can be changed in **MBASIC** by changing the value of the I/O byte (03H). This is the INTEL standard and works as follows: The byte is broken into 4 fields of two bits each.

MSD			LSD
LIST	PUNCH	READER	CONSOLE
6,7	4,5	2,3	0,1

Since these are two bits per field, they can assume the values of 0-3, and since the question from Ed is for the LST device, I'll only deal with that, the rest follow in a similar manner. The following chart shows the mapping that will occur.

LIST	DEVICE	
7	6	bits
0	0	LST mapped to TTY: (320)
0	1	LST mapped to CRT:
1	0	LST mapped to LPT: (340)
1	1	LST mapped to UL1:

Now simply **POKE** these values into 003 Hex from **MBASIC** and you can change the LST: device from within a program. For example. Assume the I/O byte contains hex value 95, which it probably will for LST: = LPT:. This means 1,0 is set in bits 7,6 respectively. To change that mapping to TTY:, which is usually port 320 (for Magnolia CP/M anyway), bytes 7,6 would have to be set to 0, 0. For this discussion I am assuming the bits in the I/O byte are set as follows:

1	0	0	1	0	1	0	1	for LPT:
0	0	0	1	0	1	0	1	for TTY:

Thus the Hex value for LPT: as shown above is 95 (149 decimal), and for TTY: as shown above is 15H (21 decimal). This is a little confusing at first to figure it out, but once you know the proper values, simply **POKE** them in to change from one to another.

I realized I got a bit carried away here, and this should have been covered in Technical Forum, but I did not expect to get carried when I started on this.--ed.

Ed also has experienced problems on his H89 display similar to those of Rod Nelson (July pg 9). When something is written on the screen it sometimes may change after it has been written. When using **PIE** and scrolling it off and back on the screen, it is correct. The difference, is Rod's problem changed once on screen, and Ed's never changes while on, but is written on incorrectly at times. Rod contacted me and told me they finally fixed his. It was a bad capacitor on the video logic board. Anyone have any ideas for Ed, please contact him.

* David Orosz/ 1251 Canton Dr/ Milpitas, CA 95035 is

looking for Diablo multi-strike film ribbon reloads (internals only). He has seen them for about \$1 in quantities of 10 or more, but does not remember where he saw this. The purchaser must himself, break apart the cartridge and replace the film ribbon. Anyone familiar with this, contact Dave.

* Gordon Richards/ POB 7211A/ Orlando, FL 32854 saw an interesting book on CP/M at his local HUG meeting. He says it looked very good and appeared to be very thorough. He has not been able to locate it. Any help? Title is **Inside CP/M, A Guide for Users and Programmers with CP/M-86 and MP/M2** by David E. Cortesi, published by Holt, Rinehart and Winston (1982) ISBN 0-03-059558-4.

* Henry Beechhold/ 13 Perry Dr/ W. Trenton, NJ 08628 wants to know if anybody has had a problem getting CP/M 2.2.03 to behave properly with the DG-80 CPU board. He says his DG-80 works fine with HDOS 2.0 and CP/M 2.2.02, but not under 2.2.03. He has written to DG, with no response yet. D-G called me and told me there is a new ROM which should correct these problems.

* Fred Rose, MD/ 4206 Ruby Pl/ Bellingham, WA 98226 has the Tandon's running on the H17 controller, single density, double track and double side. He finds that some programs will not work with these drives. Two in particular are CHGVOL and BINIMAGE. When executing CHGVOL to attempt to change a system volume number, the system dies using DKH17. Yes, this volume changing program, some CLIST type programs and more were written specifically for the 40 track drives. Fred wants to know if anyone has a program that will change the volume number of the Tandon type format disks.

TECHNICAL FORUM

DISK DRIVE (catseye) ALIGNMENT

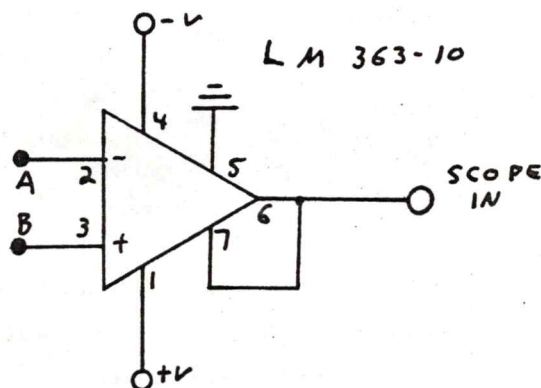
The purpose of this article is not to step you through the complete disk drive alignment procedure, but to present an easy and useful circuit you can use with an ordinary single trace scope to observe catseye patterns for alignment disk drives, and to present other hints. I felt this article was justified by the large number of disk drives in use today, and being mechanical in nature, they are likely to cause the most problems. Most of the procedures for alignment of a drive are not too complex if one has a scope, alignment disk (refer to BITS & PIECES), a disk exerciser (HS-012 offer works very well for this), some technical knowledge, and patience.

The complete procedures for alignment are found in the maintenance manuals for drives, and will not be repeated here. A sign which indicates the necessity of disk drive alignment is a diskette created in one drive (possibly by someone else) will not be readable in a second identical drive. The first thing you will note from the maintenance manual procedure is you need a differential input on the oscilloscope. This is not common to find on most scopes, especially on the not less expensive ones (under \$1000). Basically, you connect the scope inputs to the output amplifier points of the heads, with one of them inverted, and compare them differentially (A-B, with A inverted) to observe a standard catseye pattern place on the alignment diskette. This is usually found on track 16 for a 40 track drive, and track 32 for an 80 track drive. The disk drive exerciser allows you to position the head on any track and read the data, among other things.

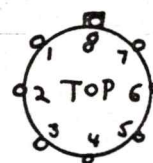
I have a fairly good dual trace scope, but it was lacking the differential input. Extended Technologies Systems

provided me with a simple circuit which I could place inside my scope or connect externally to obtain this signal conditioning, and use with one input of the scope. This circuit uses the National high performance instrument op amp, LM 363-10. It is biased with a plus and minus 9 through 15 volts. I first connected it externally to the scope using two 9V transistor batteries, and then because of the wires and mess all over the place, decided to install it in the scope, with a simple switch that could connect the output of the circuit (pin 6) to the input of the scope. I mounted a SPST switch on the front panel plus two RCA phono jacks for the inputs. I then found plus and minus 15V inside the scope and connected this directly to the op amp. The whole procedure took about an hour, and works very well. Since no other components are required with this circuit, it was very easy to install and fit inside the scope.

SCHEMATIC DIAGRAM



PINOUT

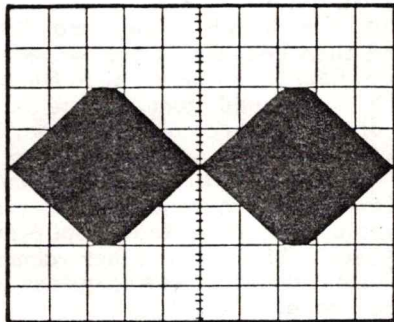


To use this circuit, you simply find the output amp points as specified in the maintenance manual. If your scope has an external trigger feature available, this is connected to the drive also in the specified place. You don't absolutely need the external trigger, but it keeps the pattern in one location for easy viewing and comparison. Since the time rate of the scope is very slow, 20 ms/DIV, there is much flickering of the display unless you have a better storage scope, long persistent phosphor scope, or memory scope.

The drive is selected and stepped to the appropriate track, and the catseye pattern as shown below is observed. If the adjacent patterns are of equal amplitude, or near (within 75%), all is well and no alignment is required (top photo). If the amplitudes are not within 75% of each other (bottom photo), alignment is required. This can cause problems. If the drive is not far off, the head carriage assembly can be loosened and a "cam screw" can be adjusted to bring the pattern within tolerance. Usually when all is tightened up again, the pattern changes, so you have to watch out. After alignment you should step the drive to track 0 and then back to the catseye position again and view, then step it out to one of the inner tracks and back to the catseye track and again view. If the pattern is stable, and

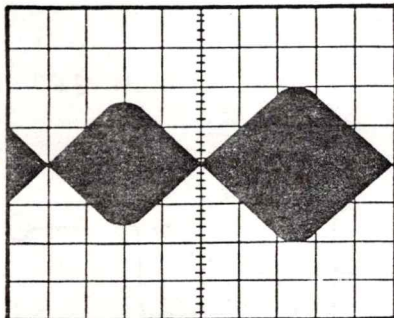
within 75%, you're done, if not, you have to do it again.

catseye pattern



EQUAL AMPLITUDE
(ON TRACK 16)

TIME SCALE - 20 ms/DIV



ONE IS 75% OF THE OTHER
(1 MIL OFF TRACK)

On the average, this takes about 15 minutes to 1/2 hour, although I have spend 2 to 3 hours on a drive already. If the head is really off, the catseye will be observed from one to 6 tracks away. Then you will have to either adjust the track 00 sensor, and/or move the entire head carriage assembly manually before the fine "cam screw" adjustment is made. For a Tandon TM-100-4, you may find the catseye pattern "off track" in multitudes of 4, since the stepper motor is a 4 phase stepper motor. Then you want to first find and align the catseye, adjust the track 00 sensor, step to 00, back to the correct catseye track, and repeat until you "find it" and it's locked in. Fortunately, most drives are not off this far and do not require as much. It is important to note, however, that the catseye alignment, the track 00 sensor, and track 00 stop are all interactive.

How does a drive know where a track is? The drive does not know where it is stepping. When access a drive, the software in use (through device drivers or BIOS) first must home the drive (find track 00), and then step out from there the number of tracks the directory information tells it something is sitting. Thus you can see if the track 00 sensor is off, all future references will be off as well. The drive sends a signal back to the disk controller when track 00 has been detected. This is passed to the system where the software can monitor it.

The track 00 stop is equally important. Its function is to prevent the drive head from going past track 00, as in many cases, this will cause the drive to hang up, and the only solution is to manually push the head past the 00 sensor again (with power off please). True the head stops when track 00 sensor is reached, but if it had a long way to travel to get there, inertia may cause it to continue a track or so past the point where the sensor tells the system "I detected track 00, stop stepping out". On the other hand, if the track 00 stop is turned in to far, the head will not be able to get properly lined up with track 00, since it will be stopped before it reaches the sensor.

There are many other signals to look for and adjustments to make for track 00 sensor, track 00 stop and others, but these will not be covered here. The most important is the catseye, as this is what usually is off and causes the most problems. If you looked at the pattern shown here, you will know why it is called a catseye pattern. And now you know the rest of the story.--Henry Fale

CLASSIFIEDS

FOR SALE-- H-14 printer in excellent condition, \$250 or best offer. Michael Couch/ 1218 18th St./ W. Des Moines, IA 50265/ (515) 223-0549.

WANTED-- Dual H17 with controller card and HDOS system. Must be 100% functional. Contact Bernard Helms/ 1565 Almond/ Roseburg, OR 97470/ (503) 673-5068. No calls between 6PM PDT Friday eve and 7AM Sunday please.

FOR SALE-- Two DG-32D 32K RAM cards for H8 computer. \$175 each or best offer. Heath Extended configuration option, \$25 or best offer. Heath H8 8080 CPU card make offer. Rick Indiano/ 11 Terry Ln/ Rochester, NY 14624.

WANTED-- Bare H8 with 8080 CPU and Front Panel card for a reasonable price. Contact George Weitzman/ 580 NW 99Th Way/ Tempbrooke Pines, FL 33024/ (305) 431-4043 after 6PM EST.

FOR SALE-- Complete H8 system including 64K Trionyx Memory, H9 terminal, H8-5 Serial board, 3 drive H-17 disk system, Epson MX-80 printer, IBM Mag Tape Selectric Typewriter in a desk converted to run from the H8, and IBM 029 card punch. \$4000 or reasonable offer. Also has CP/M and MBASIC programs available. Carroll Waddell/ 114 Dorcel/ Sumter, SC 29150.

Carroll also has written several programs to use the system as a typewriter. These programs in MBASIC allow the creation of a sequential file which can be viewed and altered, then printed to any physical printer called by its logical name in CP/M. When printed, the line length can be specified, and lines can be automatically centered in the page. These are simple menu driven programs, and provide the ability to use the terminal like a typewriter. They are not similar to word processing programs, the price of \$18 will make that clear. Programs are supplied on 5" hard sector disk, or program source code is available.

WANTED-- H17 disk controller card for the H8, and Trionyx 64K RAM card. Will pay a reasonable price. Tom Rule/ 6 Dix Terrace/ Winchester, MA 01890/ (617) 271-3282 office, 729-0298 home.

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